

2106.07890: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2106.07890_E00004	7	■ 0.255	$\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$	$\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$	$\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$
2106.07890_E00012	11	■ 0.403	$\begin{aligned} & \mathcal{L}(\text{red,red firetruck}) = 0.02 \\ & \mathcal{L}(\text{red,red idea}) = 10^{-5} \\ & \mathcal{L}(\text{red, blue sky}) = 0 \end{aligned}$	$\begin{aligned} \mathcal{L}(\text{red,red firetruck}) &= 0.02 \\ \mathcal{L}(\text{red,red idea}) &= 10^{-5} \\ \mathcal{L}(\text{red, blue sky}) &= 0 \end{aligned}$	$\begin{aligned} \mathcal{L}(\text{red,red firetruck}) &= 0.02 \\ \mathcal{L}(\text{red,red idea}) &= 10^{-5} \\ \mathcal{L}(\text{red,blue sky}) &= 0. \end{aligned}$
2106.07890_E00030	18	■ 0.508	$\begin{aligned} & \mathcal{L}(c, d) \lim_W F(c) = \min \left\{ \frac{\mathcal{L}(c, d) f(c)}{w_1}, \frac{\mathcal{L}(c, d) g(c)}{w_2}, \mathcal{L}(c, d) \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & = \lim_W F(d) \end{aligned}$	$\begin{aligned} \mathcal{L}(c, d) \lim_W F(c) &= \min \left\{ \frac{\mathcal{L}(c, d) f(c)}{w_1}, \frac{\mathcal{L}(c, d) g(c)}{w_2}, \mathcal{L}(c, d) \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & = \lim_W F(d), \end{aligned}$	$\begin{aligned} \mathcal{L}(c, d) \lim_W F(c) &= \min \left\{ \frac{\mathcal{L}(c, d) f(c)}{w_1}, \frac{\mathcal{L}(c, d) g(c)}{w_2}, \mathcal{L}(c, d) \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & = \lim_W F(d), \end{aligned}$
2106.07890_E00024 (16)	17	■ 0.538	$\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$	$\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$	$\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$

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2106.07890_E00034	19	0.573	$\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) &= \inf_{c \in \mathcal{L}} \{[hc, (f \times g)c]\} \\ &= \inf_{c \in \mathcal{L}} \{[hc, \min\{fc, gc\}]\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \min_{\left\{ \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, 1 \right\}, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{hc}, 1 \right\} \right\}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$	$\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) &= \inf_{c \in \mathcal{L}} \{[hc, (f \times g)c]\} \\ &= \inf_{c \in \mathcal{L}} \{[hc, \min\{fc, gc\}]\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \min_{\left\{ \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, 1 \right\}, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{hc}, 1 \right\} \right\}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$	
2106.07890_E00044	21	0.657	$\begin{aligned} \widehat{\mathcal{L}}([h^x, h^y](c)) &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$	$\begin{aligned} [h^x, h^y](c) &= \widehat{\mathcal{L}}(h^c \times h^x, h^y) \\ &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$	$\begin{aligned} [h^x, h^y](c) &= \widehat{\mathcal{L}}(h^c \times h^x, h^y) \\ &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$
2106.07890_E00003 (2)	7	0.667	$\left\{ (h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d) \right\}_{d \in \text{ob}(\mathcal{C})}$	$\left\{ (h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d) \right\}_{d \in \text{ob}(\mathcal{C})}$	$\left\{ (h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d) \right\}_{d \in \text{ob}(\mathcal{C})}$
2106.07890_E00020 (12)	14	0.776	$c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise} \end{cases}$	$c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise} \end{cases}$	$c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise.} \end{cases}$
2106.07890_E00032 (19)	18	0.783	$\left((w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise} \end{cases}$	$\left((w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise} \end{cases}$	$\left((w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise.} \end{cases}$
2106.07890_E00005	7	0.797	$\left[h^x, h^y \right](c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise} \end{cases}$	$\left[h^x, h^y \right](c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise} \end{cases}$	$\left[h^x, h^y \right](c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise.} \end{cases}$

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2106.07890_E00008 (6)	10	■ 0.890	[a, b]:= \begin{cases}b / a & \text { if } b<a \\ 1 & \text { otherwise } \end{cases} .\end{cases}	$[a, b] := \begin{cases} b/a & \text{if } b < a \\ 1 & \text{otherwise} \end{cases} .$	$[a, b] := \begin{cases} b/a & \text{if } b < a, \\ 1 & \text{otherwise.} \end{cases}$
2106.07890_E00040	20	■ 0.955	\begin{aligned} & \operatorname{colim}_W F(c) = \max \left\{ \mathcal{L}(c, d) w_1 f(c), \mathcal{L}(c, d) w_2 g(c) \right\} \\ & \leq \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\} \\ & = \operatorname{colim}_W F(d), \end{aligned}	$\mathcal{L}(c, d) \operatorname{colim}_W F(c) = \max \{ \mathcal{L}(c, d) w_1 f(c), \mathcal{L}(c, d) w_2 g(c) \}$ $\leq \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\}$ $= \operatorname{colim}_W F(d),$	$\mathcal{L}(c, d) \operatorname{colim}_W F(c) = \max \{ \mathcal{L}(c, d) w_1 f(c), \mathcal{L}(c, d) w_2 g(c) \}$ $\leq \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\}$ $= \operatorname{colim}_W F(d),$
2106.07890_E00048	23	■ 0.967	\begin{aligned} & \left((w_1, f) \times (w_2, g) \right) (c) = -\ln \left(\left((w'_1, f') \times (w'_2, g') \right) (c) \right) \\ & = -\ln \left(\min \left\{ \frac{f'c}{w'_1}, \frac{g'c}{w'_2}, 1 \right\} \right) \\ & = \max \left\{ -\ln \left(\frac{f'c}{w'_1} \right), -\ln \left(\frac{g'c}{w'_2} \right), 0 \right\} \\ & = \max \{ -\ln(f'c) + \ln(w'_1), -\ln(g'c) + \ln(w'_2), 0 \} \\ & = \max \{ fc - w_1, gc - w_2, 0 \}. \end{aligned}	$((w_1, f) \times (w_2, g))(c) = -\ln(((w'_1, f') \times (w'_2, g'))(c))$ $= -\ln\left(\min\left\{\frac{f'c}{w'_1}, \frac{g'c}{w'_2}, 1\right\}\right)$ $= \max\left\{-\ln\left(\frac{f'c}{w'_1}\right), -\ln\left(\frac{g'c}{w'_2}\right), 0\right\}$ $= \max\{-\ln(f'c) + \ln(w'_1), -\ln(g'c) + \ln(w'_2), 0\}$ $= \max\{fc - w_1, gc - w_2, 0\}$	$((w_1, f) \times (w_2, g))(c) = -\ln(((w'_1, f') \times (w'_2, g'))(c))$ $= -\ln\left(\min\left\{\frac{f'c}{w'_1}, \frac{g'c}{w'_2}, 1\right\}\right)$ $= \max\left\{-\ln\left(\frac{f'c}{w'_1}\right), -\ln\left(\frac{g'c}{w'_2}\right), 0\right\}$ $= \max\{-\ln(f'c) + \ln(w'_1), -\ln(g'c) + \ln(w'_2), 0\}$ $= \max\{fc - w_1, gc - w_2, 0\}.$
2106.07890_E00029	17	■ 0.990	\begin{aligned} & [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}. \end{aligned}	$[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}.$	$[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}.$

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2106.07890_E00037	19	■ 0.996	$\begin{aligned} & \widehat{\operatorname{colim}_W F i, Z} = \inf_{c \in \mathcal{L}} \{ [\max \{w_1 f c, w_2 g c\}, Z c] \} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{w_1 f c, w_2 g c\}}, 1 \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}. \end{aligned}$	$\begin{aligned} \widehat{\mathcal{L}}(\operatorname{colim}_W F i, Z) &= \inf_{c \in \mathcal{L}} \{ [\max \{w_1 f c, w_2 g c\}, Z c] \} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{w_1 f c, w_2 g c\}}, 1 \right\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}. \end{aligned}$	$\begin{aligned} \widehat{\mathcal{L}}(\operatorname{colim}_W F i, Z) &= \inf_{c \in \mathcal{L}} \{ [\max \{w_1 f c, w_2 g c\}, Z c] \} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{w_1 f c, w_2 g c\}}, 1 \right\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}. \end{aligned}$
2106.07890_E00039	20	■ 0.997	$\begin{aligned} & \{ [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) \} = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}. \end{aligned}$	$\begin{aligned} [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) &= \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}. \end{aligned}$	$\begin{aligned} [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) &= \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}. \end{aligned}$
2106.07890_E00035	19	■ 0.999	$\left(w_1, f \right) \sqcup \left(w_2, g \right) := \operatorname{colim}_W F.$	$(w_1, f) \sqcup (w_2, g) := \operatorname{colim}_W F.$	$(w_1, f) \sqcup (w_2, g) := \operatorname{colim}_W F.$
2106.07890_E00001	7	■ 1.000	$\left[h^{\text{red}}, h^{\text{blue}} \right] : \mathbf{C} \rightarrow \mathbf{Set}$	$\left[h^{\text{red}}, h^{\text{blue}} \right] : \mathbf{C} \rightarrow \mathbf{Set}$	$[h^{\text{red}}, h^{\text{blue}}] : \mathbf{C} \rightarrow \mathbf{Set}$
2106.07890_E00010	10	■ 1.000	$\begin{aligned} & a \times b := \min \{a, b\} \\ & a \sqcup b := \max \{a, b\}. \end{aligned}$	$\begin{aligned} a \times b &:= \min \{a, b\} \\ a \sqcup b &:= \max \{a, b\}. \end{aligned}$	$\begin{aligned} a \times b &:= \min \{a, b\} \\ a \sqcup b &:= \max \{a, b\}. \end{aligned}$
2106.07890_E00002 (1)	7	■ 1.000	$\left(h^{\text{red}}, h^{\text{blue}} \right) (c) = \widehat{\mathcal{C}}(h^c, [h^{\text{red}}, h^{\text{blue}}]) = \widehat{\mathcal{C}}(h^c \times h^{\text{red}}, h^{\text{blue}}).$	$\left[h^{\text{red}}, h^{\text{blue}} \right] (c) = \widehat{\mathcal{C}}(h^c, [h^{\text{red}}, h^{\text{blue}}]) = \widehat{\mathcal{C}}(h^c \times h^{\text{red}}, h^{\text{blue}}).$	$[h^{\text{red}}, h^{\text{blue}}](c) = \widehat{\mathcal{C}}(h^c, [h^{\text{red}}, h^{\text{blue}}]) = \widehat{\mathcal{C}}(h^c \times h^{\text{red}}, h^{\text{blue}}).$
2106.07890_E00006 (4)	9	■ 1.000	$\begin{aligned} & 1 \leq \mathcal{C}(x, x) \\ & \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{aligned}$	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$	$\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$
2106.07890_E00007 (5)	9	■ 1.000	$x \otimes y \leq z \text{ if and only if } x \leq [y, z]$	$x \otimes y \leq z \text{ if and only if } x \leq [y, z]$	$x \otimes y \leq z \text{ if and only if } x \leq [y, z]$
2106.07890_E00009 (7)	10	■ 1.000	$a b \leq c \text{ if and only if } a \leq [b, c]$	$ab \leq c \text{ if and only if } a \leq [b, c]$	$ab \leq c \text{ if and only if } a \leq [b, c]$
2106.07890_E00011	11	■ 1.000	$\mathcal{L}(x, y) := \pi(y \mid x),$	$\mathcal{L}(x, y) := \pi(y \mid x),$	$\mathcal{L}(x, y) := \pi(y \mid x),$
2106.07890_E00013 (8)	11	■ 1.000	$\pi(z \mid y) \pi(y \mid x) = \pi(z \mid x)$	$\pi(z \mid y) \pi(y \mid x) = \pi(z \mid x)$	$\pi(z \mid y) \pi(y \mid x) = \pi(z \mid x)$
2106.07890_E00014 (9)	12	■ 1.000	$\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$	$\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$	$\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$
2106.07890_E00015 (10)	12	■ 1.000	$\mathcal{D}^{\mathcal{C}}(f, g) := \int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$	$\mathcal{D}^{\mathcal{C}}(f, g) := \int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$	$\mathcal{D}^{\mathcal{C}}(f, g) := \int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$

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2106.07890_E00016 (11)	13	■ 1.000	$\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{[fc, gc]\}$	$\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{[fc, gc]\}$	$\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{[fc, gc]\}$
2106.07890_E00017	13	■ 1.000	$h^x := \mathcal{C}(x, -)$	$h^x := \mathcal{C}(x, -)$	$h^x := \mathcal{C}(x, -)$
2106.07890_E00018	13	■ 1.000	$\mathcal{C}(c, d) \leq [h^x(c), h^x(d)]$	$\mathcal{C}(c, d) \leq [h^x(c), h^x(d)]$	$\mathcal{C}(c, d) \leq [h^x(c), h^x(d)]$
2106.07890_E00019	13	■ 1.000	$\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), fx] = [1, fx] = fx$	$\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), fx] = [1, fx] = fx$	$\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), fx] = [1, fx] = fx$
2106.07890_E00021	15	■ 1.000	$x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$	$x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$	$x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$
2106.07890_E00022 (13)	16	■ 1.000	$\mathrm{C}(-, \lim F) \cong \operatorname{Set}^{\mathbf{J}}(*, \mathrm{C}(-, F))$	$\mathrm{C}(-, \lim F) \cong \operatorname{Set}^{\mathbf{J}}(*, \mathrm{C}(-, F))$	$\mathrm{C}(-, \lim F) \cong \operatorname{Set}^{\mathbf{J}}(*, \mathrm{C}(-, F))$
2106.07890_E00023 (15)	16	■ 1.000	$\mathcal{E}(-, \lim_W F) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$	$\mathcal{E}(-, \lim_W F) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$	$\mathcal{E}(-, \lim_W F) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$
2106.07890_E00025	17	■ 1.000	$(w_1, f) \times (w_2, g) := \lim_W F$	$(w_1, f) \times (w_2, g) := \lim_W F$	$(w_1, f) \times (w_2, g) := \lim_W F$
2106.07890_E00026 (17)	17	■ 1.000	$\widehat{\mathcal{L}}(Z, \lim_W F) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F))$	$\widehat{\mathcal{L}}(Z, \lim_W F) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F))$	$\widehat{\mathcal{L}}(Z, \lim_W F) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F))$
2106.07890_E00027	17	■ 1.000	$\inf_{c \in \mathcal{L}} \left\{ \left[Zc, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}$	$\inf_{c \in \mathcal{L}} \left\{ \left[Zc, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}$	$\inf_{c \in \mathcal{L}} \left\{ \left[Zc, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}$
2106.07890_E00028	17	■ 1.000	$\widehat{\mathcal{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{[Zc, Fic]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$	$\widehat{\mathcal{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{[Zc, Fic]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$	$\widehat{\mathcal{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{[Zc, Fic]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$
2106.07890_E00031 (18)	18	■ 1.000	$((w_1, h^x) \times (w_2, h^y))(c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$	$((w_1, h^x) \times (w_2, h^y))(c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$	$((w_1, h^x) \times (w_2, h^y))(c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$
2106.07890_E00033	18	■ 1.000	$\widehat{\mathcal{L}}(h, f \times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g)$	$\widehat{\mathcal{L}}(h, f \times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g)$	$\widehat{\mathcal{L}}(h, f \times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g)$
2106.07890_E00036 (20)	19	■ 1.000	$\widehat{\mathcal{L}}(\operatorname{colim}_W F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z))$	$\widehat{\mathcal{L}}(\operatorname{colim}_W F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z))$	$\widehat{\mathcal{L}}(\operatorname{colim}_W F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z))$
2106.07890_E00038	19	■ 1.000	$\widehat{\mathcal{L}}(Fi, Z) = \inf_{c \in \mathcal{L}} \{[Fic, Zc]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Zc}{Fic}, 1 \right\}$	$\widehat{\mathcal{L}}(Fi, Z) = \inf_{c \in \mathcal{L}} \{[Fic, Zc]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Zc}{Fic}, 1 \right\}$	$\widehat{\mathcal{L}}(Fi, Z) = \inf_{c \in \mathcal{L}} \{[Fic, Zc]\} = \inf_{c \in \mathcal{L}} \left\{ \frac{Zc}{Fic}, 1 \right\}$
2106.07890_E00041	20	■ 1.000	$[f, g](c) := \widehat{\mathcal{L}}(h^c \times f, g)$	$[f, g](c) := \widehat{\mathcal{L}}(h^c \times f, g)$	$[f, g](c) := \widehat{\mathcal{L}}(h^c \times f, g)$
2106.07890_E00042	21	■ 1.000	$\widehat{\mathcal{L}}(h \times f, g) = \widehat{\mathcal{L}}(h, [f, g])$	$\widehat{\mathcal{L}}(h \times f, g) = \widehat{\mathcal{L}}(h, [f, g])$	$\widehat{\mathcal{L}}(h \times f, g) = \widehat{\mathcal{L}}(h, [f, g])$
2106.07890_E00043 (21)	21	■ 1.000	$[h^x, h^y](c) = \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}$	$[h^x, h^y](c) = \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}$	$[h^x, h^y](c) = \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}$

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2106.07890_E00045	23	■ 1.000	$\widehat{\mathcal{M}}(f, g) = \sup_{x \in \mathcal{M}} d_{[0, \infty]}(fx, gx) = \sup_{x \in \mathcal{M}} \max\{gx - fx, 0\}$		
2106.07890_E00046	23	■ 1.000	$\begin{gathered} \left(\left(\left(w_1, f \right) \times \left(w_2, g \right) \right) \right) (c) = \max \left\{ fc - w_1, gc - w_2, 0 \right\} \\ \left(\left(w_1, f \right) \sqcup \left(w_2, g \right) \right) (c) = \min \left\{ fc + w_1, gc + w_2 \right\} \end{gathered}$	$\begin{aligned} ((w_1, f) \times (w_2, g))(c) &= \max\{fc - w_1, gc - w_2, 0\} \\ ((w_1, f) \sqcup (w_2, g))(c) &= \min\{fc + w_1, gc + w_2\} \end{aligned}$	$\begin{aligned} ((w_1, f) \times (w_2, g))(c) &= \max\{fc - w_1, gc - w_2, 0\} \\ ((w_1, f) \sqcup (w_2, g))(c) &= \min\{fc + w_1, gc + w_2\} \end{aligned}$
2106.07890_E00047	23	■ 1.000	$\left(\left(\left(w_1^{\prime}, f^{\prime} \right) \times \left(w_2^{\prime}, g^{\prime} \right) \right) \right) (c) = \min \left\{ \frac{f^{\prime} c}{w_1^{\prime}}, \frac{g^{\prime} c}{w_2^{\prime}}, 1 \right\}$	$((w_1', f') \times (w_2', g'))(c) = \min \left\{ \frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1 \right\}$	$((w_1', f') \times (w_2', g'))(c) = \min \left\{ \frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1 \right\}$
2106.07890_E00049	24	■ 1.000	$\begin{aligned} &\left(\left(\left(w_1, f \right) \times \left(w_2, g \right) \right) \right) (c) \text{ and } \left(\left(\left(w_1^{\prime}, f^{\prime} \right) \times \left(w_2^{\prime}, g^{\prime} \right) \right) \right) (c) \\ &\text{and } \left(\left(\left(w_1, f \right) \sqcup \left(w_2, g \right) \right) \right) (c) \text{ and } \left(\left(\left(w_1^{\prime}, f^{\prime} \right) \sqcup \left(w_2^{\prime}, g^{\prime} \right) \right) \right) (c) \\ &= \min \left\{ \left(\left(\left(w_1, f \right) \times \left(w_2, g \right) \right) \right) (c), \left(\left(\left(w_1^{\prime}, f^{\prime} \right) \times \left(w_2^{\prime}, g^{\prime} \right) \right) \right) (c), \right. \\ &\quad \left. \left(\left(\left(w_1, f \right) \sqcup \left(w_2, g \right) \right) \right) (c), \left(\left(\left(w_1^{\prime}, f^{\prime} \right) \sqcup \left(w_2^{\prime}, g^{\prime} \right) \right) \right) (c) \right\} \end{aligned}$	$\begin{aligned} ((w_1, f) \sqcup (w_2, g))(c) &= -\ln(((w_1', f') \sqcup (w_2', g'))(c)) \\ &= -\ln(\max\{w_1' f' c, w_2' g' c\}) \\ &= \min\{-\ln(w_1' f' c), -\ln(w_2' g' c)\} \\ &= \min\{-\ln(f' c) - \ln(w_1'), -\ln(g' c) - \ln(w_2')\} \\ &= \min\{fc + w_1, gc + w_2\}. \end{aligned}$	$\begin{aligned} ((w_1, f) \sqcup (w_2, g))(c) &= -\ln(((w_1', f') \sqcup (w_2', g'))(c)) \\ &= -\ln(\max\{w_1' f' c, w_2' g' c\}) \\ &= \min\{-\ln(w_1' f' c), -\ln(w_2' g' c)\} \\ &= \min\{-\ln(f' c) - \ln(w_1'), -\ln(g' c) - \ln(w_2')\} \\ &= \min\{fc + w_1, gc + w_2\}. \end{aligned}$
2106.07890_E00050	24	■ 1.000	$s_1 \oplus s_2 = \min\{s_1, s_2\} \text{ and } s_1 \odot s_2 = s_1 + s_2$	$s_1 \oplus s_2 = \min\{s_1, s_2\} \text{ and } s_1 \odot s_2 = s_1 + s_2$	$s_1 \oplus s_2 = \min\{s_1, s_2\} \text{ and } s_1 \odot s_2 = s_1 + s_2$
2106.07890_E00051	25	■ 1.000	$\left(\left(\left(w_1, f \right) \times \left(w_2, g \right) \right) \right) (c) = \min\{fc + w_1, gc + w_2\} = w_1 \odot fc \oplus w_2 \odot gc.$	$((w_1, f) \sqcup (w_2, g))(c) = \min\{fc + w_1, gc + w_2\} = w_1 \odot fc \oplus w_2 \odot gc.$	$((w_1, f) \sqcup (w_2, g))(c) = \min\{fc + w_1, gc + w_2\} = w_1 \odot fc \oplus w_2 \odot gc.$